Yi Yang

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Research interests: Robotic manipulation; deformable object modeling; dexterous manipulation.

Education

Shanghai Jiao Tong University Sep. 2023 – Jun. 2027 (expected)
B.Eng. in Engineering Mechanics; Zhiyuan Honors Program *Shanghai, China*

- Average grade: 89/100.
- Selected coursework: Mathematical Analysis (Honors), Theoretical Mechanics (Honors), Robot Tactile Perception.

University of California, Berkeley Jun. – Aug. 2024
Summer Session *Berkeley, CA*

- Artificial Intelligence (A+); Discrete Mathematics (A). GPA: 4.0.

Publications

* and † mark equal contribution within each group.

Point It, Strike It: Direction-Conditioned Dynamic Manipulation of Deformable Linear Objects
Yi Yang*, Xiang Fei*, Lehong Wang*, Zilin Dai†, Ruogu Li†, Jiting Cai†, Liyao Chang, Xinyi Yang, Henry Kou, Ruijie Fu, Lu Li, Howie Choset.
Preprint, 2026. [Paper] [Project]

DeformX: A Versatile Co-Simulation Framework for Deformable Linear Objects
Yi Yang*, Xiang Fei*, Lehong Wang*, Chenhao Li, Zilin Dai, Henry Kou, Lu Li, Howie Choset.
IEEE/RSJ IROS 2026. Accepted. [Paper] [Project]

ENPIRE: Agentic Robot Policy Self-Improvement in the Real World
Wenli Xiao*, Jia Xie*, Tonghe Zhang*, Haotian Lin*, Letian "Max" Fu, Haoru Xue, Jalen Lu, **Yi Yang**, Cunxi Dai, Zi Wang, Jimmy Wu, Guanzhi Wang, S. Shankar Sastry, Ken Goldberg, Linxi "Jim" Fan, Yuke Zhu, Guanya Shi.
Conference on Robot Learning (CoRL), 2026. Accepted. [Paper] [Project]

Development of a 15-Degree-of-Freedom Bionic Hand with Cable-Driven Transmission and Distributed Actuation
Haoqi Han, **Yi Yang**, Yifei Yu, Yixuan Zhou, Xiaohan Zhu, Hesheng Wang.
IEEE/ASME Transactions on Mechatronics (T-MECH), 2026. Accepted. [Paper]

Research Experience

LeCAR Lab Mar. 2026 – Present
Research Intern, Carnegie Mellon University Advisor: Prof. Guanya Shi.

- Developed the simulation and motion-planning components of *ENPIRE*, supporting the evaluation and execution of agent-generated robot manipulation policies.

Biorobotics Lab Oct. 2025 – Oct. 2026
Research Intern, Carnegie Mellon University Advisor: Prof. Howie Choset.

- Designed the overall architecture and developed the Isaac Sim integration for *DeformX*, coupling Cosserat rod dynamics with robotic simulation for synthetic data generation and policy learning.

MPI Lab Feb. 2025 – Present
Research Intern, Shanghai Jiao Tong University Advisor: Prof. Daolin Ma.

- Investigating tactile-based imitation learning for in-hand manipulation, including observation design and evaluation under variations in object pose and contact conditions.

IRMV Lab Nov. 2023 – Feb. 2025
Research Intern, Shanghai Jiao Tong University Advisor: Prof. Hesheng Wang.

- Designed the core electronic circuits and contributed to the mechanical structure design of a *15-DoF tendon-driven bionic hand*.
- *16-DoF visual dexterous manipulation*: developed the kinematic model, simulation, and mechanical prototype; integrated visual feedback, posture control, and trajectory tracking.

Honors & Awards

Zhiyuan Honorary Scholarship 2024, 2025
Second Prize Scholarship 2024
Shanghai College Student Mechanics Competition, First Prize 2024
National First Prize, National Physics Olympiad in China 2022
Provincial Second Prize, National Olympiad in Informatics in Provinces 2020